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Assessing the impact of healthcare - income maximisation models of care for families of children aged 0-5 years experiencing financial difficulties, on family finances, and parent/caregiver(s) or children's health and wellbeing.: A systematic review protocol.

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Assessing the impact of healthcare - income maximisation models of care for families of children aged 0-5 years experiencing financial difficulties, on family finances, and parent/caregiver(s) or children's health and wellbeing.: A systematic review protocol.

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Abstract

Introduction

Poverty has far-reaching and detrimental effects on children's physical and mental health, across all geographies. Financial advice and income-maximisation services can provide a promising opportunity for shifting the physical and mental health burdens that commonly occur with financial hardship, yet awareness of these services is limited, and referrals are not systematically integrated into existing healthcare service platforms. We aim to map and synthesise evidence on the impact of healthcare-income maximisation models of care for families of children aged 0-5 years in high-income countries on family finances, parent/caregiver(s) or children's health and wellbeing.

Methods and analysis

To be included in the review, studies must be (1) families (expectant mothers or parents/caregivers) of children who are aged between 0- to 5-years, accessing a healthcare service, (2) include a referral from healthcare to an income-maximisation service (i.e., financial counselling), and (3) examine impacts on child and family health and wellbeing. A comprehensive, electronic search strategy will be used to identify studies written in English, published from inception until January 2021, and indexed in MEDLINE, EMBase, PsycInfo, CINAHL, Proquest, Family & Society Studies Worldwide, Cochrane Library, and Informit Online. Search strategies will include terms for: families, financial hardship, and healthcare, in various combinations. Bibliographies of primary studies and review articles meeting the inclusion criteria will be searched manually to identify further eligible studies. Data on objective and self-reported outcomes and study quality will be independently extracted by two review authors; any disagreements will be resolved through a third reviewer. The protocol follows the Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols (PRISMA-P).

Ethics and dissemination

Ethical approval is not required. The results will be disseminated widely via peer-reviewed publication and presentations at conferences related to this field.

PROSPERO registration number CRD42020195985

Keywords

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Systematic review; income maximisation; financial counselling; child health service; parental health and wellbeing

Strengths and limitations

- To our knowledge, this is the first systematic review to explore the impact of healthcare-income maximisation models of care on family finances, parent/caregiver(s) or children's health and wellbeing.
- The protocol follows the Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols guidelines
- A comprehensive search strategy across several health research-related databases is included in this study
- Non-English language electronic databases will not be searched which is a limitation
- A potential limitation of this review may be the heterogeneity in reporting and measures of outcomes

Introduction

It is recognised that social determinants of health (SDOH: conditions in which people are born, grow, live, work, and age; 1) will impact children's health and development either directly or indirectly long term (2, 3). Childhood poverty has a damaging impact on every dimension of health, wellbeing and development (4). As children raised in poverty become adults, they experience a greater subsequent risk of educational failure, underemployment, lack of societal participation, risky behaviours and delinquency (5). It is estimated that one in five children in high-income countries lives in relative income poverty (6).

Compounding this, COVID-19 has created significant economic disruption across the globe, exacerbating and entrenching disadvantage via widespread job losses and financial stress. This has resulted in major adverse consequences on the health and well-being of families worldwide. It was estimated that between 88 and 115 million people around the globe would be pushed into extreme poverty in 2020, and this is expected to rise to between 119 and 124 million across 2021 (7). While COVID-19 is viewed as a temporary shock to economic growth, the economic damage to families that are already experiencing financial hardship

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could be more long-lasting and may increase the number of families who are now experiencing poverty.

In response to socioeconomic and health inequities, researchers and policymakers have sought to design effective interventions that reduce child poverty directly (8). Systematic reviews have found that unconditional cash transfers and increases in household cash income have been associated with beneficial effects on children's health in low-income/middle-income and high-income countries (9-11). The exact causal mechanisms underlying these effects are broadly still being explored, but are likely related to addressing material deprivation to build and invest in nurturing home environments, reduce parental stress, increase emotionally-responsive parenting, strengthen kin relationships, and increase household and family stability (12). None, however, have focused exclusively on referral pathways from healthcare to income-maximisation services.

Income-maximisation advice services such as financial counsellors and welfare advice services can enable families to access financial entitlements they may not be aware of, provide assistance with childcare and housing options, and offer support and advice in crisis situations. Embedded income maximisation services within a health care setting provides a unique opportunity to take action due to the potential for wide-scale early screening, identification and referrals (13). The literature on the impact of financial services, and particularly the healthcare-financial service interface, is a disparate field of published and grey literature which has yet to be reviewed systematically (14-19). We aim to fill this evidence gap by conducting the first systematic literature review on this topic to inform better parental/caregiver(s) or children health and wellbeing though referral from a healthcare service (i.e., provided by paediatricians, child health nurses, general practitioners, health social workers, health professionals, nurse, doctor, or midwife) to an income-maximisation service (i.e., financial counsellor, Citizens Advice Bureau).

Methods and analysis

Protocol and Registration

This systematic review will be conducted following the recommendation of the Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA-P) guidelines (20). This review protocol is registered in the International Prospective Register of Systematic Reviews (PROSPERO) database and can be accessed at: https://www.crd.york.ac.uk/prospero/display_record.php?ID=CRD42020195985

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See supplementary file 1 for the completed PRISMA-P checklist.

Aim & Design

The purpose of this systematic review is to synthesise and critically evaluate the scientific evidence on the impact of healthcare - income maximisation models of care for families of children aged 0-5 years experiencing financial difficulties on family finances and parent/caregiver(s) or children's health and wellbeing.

Search strategy

The search strategy will be performed using the Population/Intervention/Comparison/Outcomes (PICO) approach (21) to ensure a systematic search of the literature of the following research question: In families of children ages 0- to 5 years who are experiencing financial difficulties (P,0), do healthcare - income maximisation models of care (I) compared to usual care (C) have a positive impact on family finances, and/or parent/caregiver, and/or child health and wellbeing (O). Search terms will include subject headings (e.g., MeSH in PubMed/MEDLINE) in addition to free-text words (with appropriate truncation) for the key concepts in relation to the population (that is, families will include expectant mothers/parents/caregivers of children who are aged between 0- to 5-years), intervention (that is, referrals from healthcare to income maximisation services) outcome of interest (that is, financial impact). The search strategy will be developed with an experienced librarian, (see sample search strategy in supplementary file 2) and adapted to the other bibliographic databases. For general databases that do not have subject headings, free-text keyword searches will be applied. Article searches will be conducted in the following specialised and general databases from inception to January 2021: Medical Literature Analysis and Retrieval System Online (MEDLINE), Excerpta Medica database (Embase), Psychology Information (PsycINFO), and (EMCare) via OVID. Cumulative Index to Nursing and Allied Health Literature (CINAHL Complete), Proquest, and Family & Society Studies Worldwide via EBSCO. Cochrane Library via Wiley, and Informit Online via RMIT. Grey literature will also be searched. The reference sections of the included studies and cited studies will be manually searched for additional relevant studies.

Inclusion criteria

Studies will be included if they are published in English. The types of participants, studies and outcomes that were considered for inclusion are described below.

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Study Eligibility

Type of study/ characteristics

Experimental and quasi-experimental study designs including randomized controlled trials, nonrandomized controlled trials, before and after studies, mixed methods evaluations and interrupted time-series studies. In addition, analytical observational studies including prospective and retrospective cohort studies, case-control studies and analytical cross-sectional studies will be considered for inclusion. Studies must have a component of healthcare-income maximisation models of care. This includes referral pathways (i.e., a form of linkage) from healthcare services (e.g., provided by paediatricians, child health nurses, general practitioners, health social workers, health professionals, nurses, doctors, or midwives) to an income maximisation service (e.g., health referrals to financial counselling services, or co-location of health and financial counselling services). For the purposes of this review, income maximisation services may examine—but are not limited to—family's income, source of income, current debts (utility, taxation, gambling, business, debt recovery, loans, or credit cards), unemployment issues, change in circumstances (disability), budgeting and expenses (accommodation, utilities, food, or travel), and financial literacy/ financial education. For experimental studies, the interventions will need to examine an outcome based on a referral pathway between healthcare-income maximisation services. For longitudinal studies, any follow-up length is allowed but there must be at least one outcome measure. There will be no limit on sample size for study designs. Grey literature (e.g., book chapters, dissertations, conference abstracts), registered clinical trials and government reports/guidelines) will be included.

In this review, we will consider all studies conducted in English, and conducted in high income countries, where it is possible to calculate or report on financial impact, or health and wellbeing, when a child is aged between 0- to 5-years old. Studies will be ineligible if there is a focus on cash transfer programs.

Type of comparators

The studies should compare the income maximisation model to usual care. Studies comparing different type of income maximisation will be included.

Population

Families (expectant mothers or parents/caregivers) of children who are aged between 0- to 5-years who are accessing a child healthcare service. If age range is reported instead of a mean, samples with a lower limit of 1 month-59.99 months/4.99 years and an upper limit of

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<6 years will be eligible. If a mean age or age range is not reported, samples described as babies, infants, toddlers, or pre-schoolers will be eligible. For longitudinal or experimental study designs, the age criterion will be applied to the first measurement time point of the exposure, and could examine primary school children, adolescents, and adults. While the intervention studies may occur during the antenatal period, or when a child is aged between 0- to 5-years, outcomes can be measured with a lag.

Outcome of interest

The primary outcome will be to reflect on 1) financial impact - change in income/earnings/ debt management and 2) non-financial impact - financial literacy on the families.

Secondary outcomes

Parental/caregiver or child health and wellbeing, including physical, mental and spiritual health, social wellbeing, developmental wellbeing of the child, parenting skills, and service use. Experimental studies may include an economic evaluation of an intervention or potential intervention harm (e.g., family satisfaction survey).

Exclusion criteria

1) studies are conducted in low/ middle income countries and/or focus on cash transfer programs.
2) studies do not include a measure of financial impact, or health and wellbeing when a child is aged between 0- to 5-years old.
3) studies use a qualitative methodology or case series.

Study selection and data extraction

Bibliographic records will be extracted from the databases interfaces and imported into Reference Manager Software (Covidence, Melbourne, Australia) for de-duplication. In level 1 & 2, titles and then abstracts of potentially relevant articles retrieved from the database and web searches and citations of relevant studies will be screened by independent reviewers using Covidence software (a secure, internet-based software). In level 3, full text copies of articles will be obtained for those meeting initial eligibility screening. Any discrepancies will be resolved through discussion with a third reviewer. The same independent reviewers who examine titles and abstracts will examine all full text articles. Using the format of the validated standard data extraction form, we will extract the following information: descriptive study characteristics (e.g., author, publication year, study design, country, sample size, age, sex), exposure, outcome, results, and confounders. Where multiple models are reported

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(e.g., bivariate and adjusted models), results will be extracted from the most fully adjusted model. Findings will be determined to be statistically significant if $p<0.05$ is reported. Reviewers will not be blinded to the authors or journals when extracting data. A flow chart showing the studies included and excluded at each stage of the study selection process will be provided.

Risk of bias (quality) assessment among included studies

Using the GRADE (Grading of Recommendations Assessment, Development and Evaluation) framework (22), we will systematically examine the quality of primary research contributing to each outcome, reflecting the level of confidence in the estimated effects. Briefly, five assessment criteria (risk of bias, inconsistency, indirectness, imprecision, other) will be used to rate quality of evidence as “high”, “moderate”, “low”, or “very low”. Quality of evidence ratings will start at “high” for RCTs and “low” for all other experimental and observational studies. The quality of evidence could be downgraded for any study design due to limitations associated with the five assessment criteria. Risk of bias assessment for individual intervention studies will be completed following the Cochrane Handbook. GRADE does not have an official tool for assessing risk of bias in observational studies, which is one component of the quality of evidence, but does recommend the types of characteristics to examine (23). These will be used to assess the risk of bias of the studies included in our review. Study quality will not influence eligibility for inclusion. For non-randomised studies, we will utilise the Risk Of Bias In Non-randomized Studies of Interventions (ROBINS-I) assessment tool developed by Cochrane (24).

Addressing missing data

In case of missing data, significant empirical data such as screened, randomized, intention-to-treat, as-treated, and per-protocol population will be closely analysed. In cases where details are missing on study design, population, intervention, or outcomes, the authors of included studies will be contacted by email. After the first contact attempt, if no response is received, the study authors will be contacted 2 more times approximately 3 to 4 weeks apart. We will critically appraise the concerns related to the missing data and imputation methods, e.g., last observation carried forward (LOCF). The attrition rates, e.g., drop-outs, losses to follow-up, and withdrawals, will be investigated.

Data Synthesis

For outcomes of interest, a meta-analysis will be performed (if applicable) with the data that are sufficiently homogeneous in terms of statistical, clinical, and methodological

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characteristics. If studies are deemed not appropriate for meta-analysis, qualitative synthesis structured around the outcomes will be conducted and the results will be presented narratively. The outcomes will be presented as the mean differences (MDs)/standardized mean differences (SMDs) with 95% confidence intervals (CI) unless otherwise stated for continuous variables. For dichotomous data, the effect estimate is risk ratios (RRs) with 95% CI. Limitations of each study (i.e., quality assessment) will be described.

DISCUSSION

Financial security is fundamental for optimal child health and development, and family income is at the heart of this (4). Many high-income countries use social supports to increase family income and protect from poverty, including the provision of cash transfers, income support payments, and community services. Income maximisation advice services are an emerging area of interest as a possible method of supporting families in poverty, and healthcare service integration may provide a unique opportunity to take action in situations of financial crisis.

Understanding how income maximisation advice services may help families during crisis may be one avenue to assist families to access useful community resources and promote a comprehensive approach to poverty and deprivation, and consequently affect the health and wellbeing of families. Our review may therefore have valuable implications for stakeholders including children, families, healthcare professionals, financial counsellors, health policy managers, and researchers working in public health to provide a routine clinical assessment of financial hardship and refer to local services within the community.

A key strength of this review protocol is the use of multiple gold standard guidelines; the PRISMA-P, GRADE, and Cochrane frameworks. By incorporating different elements from each guideline, a solid framework and structure was created by which the research question could be answered. Additional strengths are the inclusion of a university librarian with experience conducting systematic reviews who assisted with peer-reviewing the search strategy, the application of a data extraction template, and a flexible approach to data acquisition and synthesis. A limitation of this review may be the heterogeneity in reporting and measures of outcomes. Non-English language electronic databases will not be searched which is another limitation.

DECLARATIONS

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Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare that they have no competing interests.

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Author contributions

JB and SW conceptualised and designed the protocol, drafted and reviewed the initial manuscript. All authors defined the concepts and search items. JB and AnnekaP defined the data extraction process as well as methodological appraisal of the studies and statistical analysis. All authors have approved and contributed to the final written manuscript.

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Competing interests

None declared.

Patient and public involvement

Patients and/or the public were not involved in the design, or conduct, or reporting, or dissemination plans of this research.

Patient consent for publication

Not required.

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Supplementary file 1: PRISMA-P checklist

PRISMA-P (Preferred Reporting Items for Systematic review and Meta-Analysis Protocols) 2015 checklist: recommended items to address in a systematic review protocol*

Section and topic	Item No	Checklist item	Page #
ADMINISTRATIVE INFORMATION			
Title:			
Identification	1a	Identify the report as a protocol of a systematic review	2
Update	1b	If the protocol is for an update of a previous systematic review, identify as such	NA
Registration	2	If registered, provide the name of the registry (such as PROSPERO) and registration number	2
Authors:			
Contact	3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	1
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review	8
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	NA
Support:			
Sources	5a	Indicate sources of financial or other support for the review	8
Sponsor	5b	Provide name for the review funder and/or sponsor	8
Role of sponsor or funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol	8
INTRODUCTION			
Rationale	6	Describe the rationale for the review in the context of what is already known	3
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	4
METHODS			
Eligibility criteria	8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	4
Information sources	9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	4,5
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could	4

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		be repeated	Supplementary file 2
Study records:			
Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	5
Selection process	11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	5
Data collection process	11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	5
Data items	12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	5,6
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	5
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	6
Data synthesis	15a	Describe criteria under which study data will be quantitatively synthesised	6
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I^2, Kendall's τ)	6
	15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	NA
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned	6
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)	NA
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	6

NA = Not applicable

From: Shamseer L, Moher D, Clarke M, Ghersi D, Liberati A, Petticrew M, Shekelle P, Stewart L, PRISMA-P Group. Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015: elaboration and explanation. BMJ. 2015 Jan 2;349(jan02 1):g7647.

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Search String

In families of children ages 0-to 5 years who are experiencing financial difficulties (P,0), do health care referrals to income maximisation services (I) compared to usual care (C) have a positive impact on family finances, and/or parent/caregiver, and/or child health and wellbeing (O).

Limits:

Language: English;

Age: expectant mothers/families/parents/guardians/or carers of children ages 0-to 5 years;

Record Type: academic journals; grey literature

Date: No limits.

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Connecting healthcare with income maximisation services, and their financial, health and wellbeing impacts for families with young children: A Systematic Review Protocol.

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Primary Subject	Health services research

Heading:	
Secondary Subject Heading:	Public health
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Connecting healthcare with income maximisation services, and their financial, health and wellbeing impacts for families with young children: a systematic review protocol

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Word count: 2551

Abstract

Introduction

Poverty has far-reaching and detrimental effects on children's physical and mental health, across all geographies. Financial advice and income-maximisation services can provide a promising opportunity for shifting the physical and mental health burdens that commonly occur with financial hardship, yet awareness of these services is limited, and referrals are not systematically integrated into existing healthcare service platforms. We aim to map and synthesise evidence on the impact of healthcare-income maximisation models of care for families of children aged 0-5 years in high-income countries on family finances, parent/caregiver(s) or children's health and wellbeing.

Methods and analysis

To be included in the review, studies must be (1) families (expectant mothers or parents/caregivers) of children who are aged between 0- to 5-years, accessing a healthcare service, (2) include a referral from healthcare to an income-maximisation service (i.e., financial counselling), and (3) examine impacts on child and family health and wellbeing. A comprehensive electronic search strategy will be used to identify studies written in English, published from inception until January 2021, and indexed in MEDLINE, EMBase, PsycInfo, CINAHL, Proquest, Family & Society Studies Worldwide, Cochrane Library, and Informit Online. Search strategies will include terms for: families, financial hardship, and healthcare, in various combinations. Bibliographies of primary studies and review articles meeting the inclusion criteria will be searched manually to identify further eligible studies, and grey literature will also be searched. Data on objective and self-reported outcomes and study quality will be independently extracted by two review authors; any disagreements will be resolved through a third reviewer. The protocol follows the Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols (PRISMA-P).

Ethics and dissemination

Ethical approval is not required. The results will be disseminated widely via peer-reviewed publication and presentations at conferences related to this field.

PROSPERO registration number CRD42020195985.

Keywords

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Systematic review; income maximisation; financial counselling; child health service; parental health and wellbeing

Strengths and limitations of this study

- To our knowledge, this is the first systematic review to explore the impact of healthcare-income maximisation models of care on family finances, parent/caregiver(s) or children's health and wellbeing.
- The protocol follows the Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols guidelines.
- A comprehensive search strategy across several health research-related databases is included in this study.
- Non-English language electronic databases will not be searched which is a limitation.
- A potential limitation of this review may be the heterogeneity in reporting and measures of outcomes.

Introduction

It is recognised that social determinants of health (SDOH: conditions in which people are born, grow, live, work, and age; 1) will impact children's health and development either directly or indirectly long term (2, 3). Childhood poverty has a damaging impact on every dimension of health, wellbeing and development (4). As children raised in poverty become adults, they experience a greater subsequent risk of educational failure, underemployment, lack of societal participation, risky behaviours and delinquency (5). It is estimated that one in five children in high-income countries lives in relative income poverty (6).

Compounding this, COVID-19 has created significant economic disruption across the globe, exacerbating and entrenching disadvantage via widespread job losses and financial stress. This has resulted in major adverse consequences on the health and well-being of families worldwide. It was estimated that between 88 and 115 million people around the globe would be pushed into extreme poverty in 2020, and this is expected to rise to between 119 and 124 million across 2021 (7). While COVID-19 is viewed as a temporary shock to economic growth, the economic damage to families that are already experiencing financial hardship

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could be more long-lasting and may increase the number of families who are now experiencing poverty.

In response to socioeconomic and health inequities, researchers and policymakers have sought to design effective interventions that reduce child poverty directly (8). Systematic reviews have found that unconditional cash transfers and increases in household cash income have been associated with beneficial effects on children's health in low-income/middle-income and high-income countries (9-11). The exact causal mechanisms underlying these effects are complex, however two theoretical models attempt to provide insight into the mechanisms. According to the investment model, families invest additional economic resources into their children, addressing material deprivation and improving health and developmental opportunities (12, 13). Alternatively, the family stress model describes how increasing economic resources can reduce parental stress and, in turn, increase their emotionally responsive parenting and improve the home environment(14). Broadly, it is understood that interventions to reduce household poverty will nurture children's health and development.

Income-maximisation advice services such as financial counsellors and welfare advice services can enable families to access financial entitlements they may not be aware of, provide assistance with childcare and housing options, and offer support and advice in crisis situations. Embedded income maximisation services within a health care setting provides a unique opportunity to take action due to the potential for wide-scale early screening, identification and referrals (15). The literature on the impact of financial services, and particularly the healthcare-financial service interface, is a disparate field of published and grey literature which has yet to be reviewed systematically (16-21). We aim to fill this evidence gap by conducting the first systematic literature review on this topic to inform better parental/caregiver(s) or children health and wellbeing through referral from a healthcare service (i.e., provided by paediatricians, child health nurses, general practitioners, health social workers, health professionals, nurse, doctor, or midwife) to an income-maximisation service (i.e., financial counsellor, Citizens Advice Bureau).

Methods and analysis

Protocol and Registration

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This systematic review will be conducted following the recommendation of the Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA-P) guidelines (22). This review protocol is registered in the International Prospective Register of Systematic Reviews (PROSPERO) database and can be accessed at: https://www.crd.york.ac.uk/prospero/display_record.php?ID=CRD42020195985

Aim & Design

The purpose of this systematic review is to synthesise and critically evaluate the scientific evidence on the impact of healthcare - income maximisation models of care for families of children aged 0-5 years experiencing financial difficulties on family finances and parent/caregiver(s) or children's health and wellbeing.

Search strategy

The search strategy will be performed using the Population/Intervention/Comparison/Outcomes (PICO) approach (23) to ensure a systematic search of the literature of the following research question: In families of children ages 0- to 5 years who are experiencing financial difficulties (P,0), do healthcare - income maximisation models of care (I) compared to usual care (C) have a positive impact on family finances, and/or parent/caregiver, and/or child health and wellbeing (O). Search terms will include subject headings (e.g., MeSH in PubMed/MEDLINE) in addition to free-text words (with appropriate truncation) for the key concepts in relation to the population (that is, families will include expectant mothers/parents/caregivers of children who are aged between 0- to 5-years), intervention (that is, referrals from healthcare to income maximisation services) outcome of interest (that is, financial impact). The search strategy was developed with an experienced librarian and adapted to the various bibliographic databases (see final search strategy in supplementary file 1). For general databases that do not have subject headings, free-text keyword searches will be applied. Article searches will be conducted in the following specialised and general databases from inception to January 2021: Medical Literature Analysis and Retrieval System Online (MEDLINE), Excerpta Medica database (Embase), Psychology Information (PsycINFO), and (EMCare) via OVID. Cumulative Index to Nursing and Allied Health Literature (CINAHL Complete), Proquest, and Family & Society Studies Worldwide via EBSCO. Cochrane Library via Wiley, and Informit Online via RMIT. Grey literature (including book chapters, dissertations, conference abstracts, government reports/guidelines) will also be searched. The reference sections of the included studies and cited studies will be manually searched for additional relevant studies.

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Inclusion criteria

Studies will be included if they are published in English. The types of participants, studies and outcomes that were considered for inclusion are described below.

Study Eligibility

Type of study/ characteristics

Experimental and quasi-experimental study designs including randomized controlled trials, nonrandomized controlled trials, before and after studies, mixed methods evaluations and interrupted time-series studies. In addition, analytical observational studies including prospective and retrospective cohort studies, case-control studies and analytical cross-sectional studies will be considered for inclusion. Studies must have a component of healthcare-income maximisation model of care. This includes referral pathways (i.e., a form of linkage) from healthcare services (e.g., provided by paediatricians, child health nurses, general practitioners, health social workers, health professionals, nurses, doctors, or midwives) to an income maximisation service (e.g., health referrals to financial counselling services, or co-location of health and financial counselling services). For the purposes of this review, income maximisation services may examine—but are not limited to—family's income, source of income, current debts (utility, taxation, gambling, business, debt recovery, loans, or credit cards), unemployment issues, change in circumstances (disability), budgeting and expenses (accommodation, utilities, food, or travel), and financial literacy/ financial education. For experimental studies, the interventions will need to examine an outcome based on a referral pathway between healthcare-income maximisation services. For longitudinal studies, any follow-up length is allowed but there must be at least one outcome measure. Grey literature (e.g., book chapters, dissertations, conference abstracts, government reports/guidelines)will be included.

In this review, we will consider all studies conducted in English, and conducted in high income countries, where it is possible to calculate or report on financial impact, or health and wellbeing, when a child is aged between 0- to 5-years old. Studies will be ineligible if there is a focus on cash transfer programs.

Type of comparators

The studies should compare the income maximisation model to usual care. Studies comparing different type of income maximisation will be included.

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Population

Families (expectant mothers or parents/caregivers) of children who are aged between 0- to 5-years who are accessing a child healthcare service. If age range is reported instead of a mean, samples with a lower limit of 1 month-59.99 months/4.99 years and an upper limit of <6 years will be eligible. If a mean age or age range is not reported, samples described as babies, infants, toddlers, or pre-schoolers will be eligible. For longitudinal or experimental study designs, the age criterion will be applied to the first measurement time point of the exposure, and could examine primary school children, adolescents, and adults. While the intervention studies may occur during the antenatal period, or when a child is aged between 0- to 5-years, outcomes can be measured with a lag.

Outcome of interest

Studies will have to measure at least one of the following outcomes: (1) income (change in income/earnings/ debt management); (2) other financial impact (financial literacy); (3) parental/caregiver or child health and wellbeing, including physical, mental and spiritual health, social wellbeing, developmental wellbeing of the child, parenting skills, service use, cost and harm.

Exclusion criteria

1) studies are conducted in low/ middle income countries and/or focus on cash transfer programs.
2) studies do not include a measure of financial impact, or health and wellbeing when a child is aged between 0- to 5-years old.
3) studies use a qualitative methodology or case series.

Study selection and data extraction

Bibliographic records will be extracted from the databases interfaces and imported into Reference Manager Software (Covidence, Melbourne, Australia) for de-duplication. In level 1 & 2, titles and then abstracts of potentially relevant articles retrieved from the database and web searches and citations of relevant studies will be screened by independent reviewers using Covidence software (a secure, internet-based software). In level 3, full text copies of articles will be obtained for those meeting initial eligibility screening. Any discrepancies will be resolved through discussion with a third reviewer. The same independent reviewers who examine titles and abstracts will examine all full text articles. Using the format of the

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validated standard data extraction form, we will extract the following information: descriptive study characteristics (e.g., author, publication year, study design, country, sample size, age, sex), exposure, outcome, results, and confounders. Where multiple models are reported (e.g., bivariate and adjusted models), results will be extracted from the most fully adjusted model. Findings will be determined to be statistically significant if $p<0.05$ is reported. Reviewers will not be blinded to the authors or journals when extracting data. A flow chart showing the studies included and excluded at each stage of the study selection process will be provided.

Risk of bias (quality) assessment among included studies

Using the GRADE (Grading of Recommendations Assessment, Development and Evaluation) framework (24), we will systematically examine the quality of primary research contributing to each outcome, reflecting the level of confidence in the estimated effects. Briefly, five assessment criteria (risk of bias, inconsistency, indirectness, imprecision, other) will be used to rate quality of evidence as “high”, “moderate”, “low”, or “very low”. Quality of evidence ratings will start at “high” for RCTs and “low” for all other experimental and observational studies. The quality of evidence could be downgraded for any study design due to limitations associated with the five assessment criteria. Risk of bias assessment for individual intervention studies will be completed following the Cochrane Handbook. GRADE does not have an official tool for assessing risk of bias in observational studies, which is one component of the quality of evidence, but does recommend the types of characteristics to examine (25). These will be used to assess the risk of bias of the studies included in our review. Study quality will not influence eligibility for inclusion. For non-randomised studies, we will utilise the Risk Of Bias In Non-randomized Studies of Interventions (ROBINS-I) assessment tool developed by Cochrane (26).

Addressing missing data

In case of missing data, significant empirical data such as screened, randomized, intention-to-treat, as-treated, and per-protocol population will be closely analysed. In cases where details are missing on study design, population, intervention, or outcomes, the authors of included studies will be contacted by email. After the first contact attempt, if no response is received, the study authors will be contacted 2 more times approximately 3 to 4 weeks apart. We will critically appraise the concerns related to the missing data and imputation methods, e.g., last observation carried forward (LOCF). The attrition rates, e.g., drop-outs, losses to follow-up, and withdrawals, will be investigated.

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Data Synthesis

For outcomes of interest, a meta-analysis will be performed (if applicable) with the data that are sufficiently homogeneous in terms of statistical, clinical, and methodological characteristics. If studies are deemed not appropriate for meta-analysis, qualitative synthesis structured around the outcomes will be conducted and the results will be presented narratively. The outcomes will be presented as the mean differences (MDs)/standardized mean differences (SMDs) with 95% confidence intervals (CI) unless otherwise stated for continuous variables. For dichotomous data, the effect estimate is risk ratios (RRs) with 95% CI. Limitations of each study (i.e., quality assessment) will be described.

Patient and public involvement

Patients and/or the public were not involved in the design, or conduct, or reporting, or dissemination plans of this research.

ETHICS AND DISSEMINATION

Ethical approval is not required for this systematic review. The results will be disseminated widely via peer-reviewed publication and presentations at conferences related to this field. A link to the published review will also be circulated via social media to disseminate the findings among academics and non-academics.

DISCUSSION

Financial security is fundamental for optimal child health and development, and family income is at the heart of this (4). Many high-income countries use social supports to increase family income and protect from poverty, including the provision of cash transfers, income support payments, and community services. Income maximisation advice services are an emerging area of interest as a possible method of supporting families in poverty, and healthcare service integration may provide a unique opportunity to take action in situations of financial crisis.

Understanding how income maximisation advice services may help families during crisis may be one avenue to assist families to access useful community resources and promote a comprehensive approach to poverty and deprivation, and consequently affect the health and wellbeing of families. Our review may therefore have valuable implications for stakeholders including children, families, healthcare professionals, financial counsellors, health policy

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managers, and researchers working in public health to provide a routine clinical assessment of financial hardship and refer to local services within the community.

A key strength of this review protocol is the use of multiple gold standard guidelines; the PRISMA-P, GRADE, and Cochrane frameworks. By incorporating different elements from each guideline, a solid framework and structure was created by which the research question could be answered. Additional strengths are the inclusion of a university librarian with experience conducting systematic reviews who assisted with peer-reviewing the search strategy, the application of a data extraction template, and a flexible approach to data acquisition and synthesis. A limitation of this review may be the heterogeneity in reporting and measures of outcomes. Non-English language electronic databases will not be searched which is another limitation.

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DECLARATIONS

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

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Author contributions

JB, SW and NS conceptualised and designed the protocol. All authors (JB, AP, AnnekaP, NS, AZ, VE, DCS, NatalieS, KL, RL, RG, SR, LK, SC, SG, SW) defined the concepts and search items. JB and AnnekaP defined the data extraction process as well as methodological appraisal of the studies and statistical analysis. This manuscript was initially drafted by JB, AP, AnnekaP, NS and SW. Subsequent drafts were commented on by all authors (JB, AP, AnnekaP, NS, AZ, VE, DCS, NatalieS, KL, RL, RG, SR, LK, SC, SG, SW). All authors have approved and contributed to the final written manuscript.

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Competing interests

The authors declare that they have no competing interests.

Patient consent for publication

Not required.

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Supplementary File 1. Search strategy

Search String

In families of children ages 0-to 5 years who are experiencing financial difficulties (P,0), do health care referrals to income maximisation services (I) compared to usual care (C) have a positive impact on family finances, and/or parent/caregiver, and/or child health and wellbeing (O).

Limits:
Language: English;
Age: expectant mothers/families/parents/guardians/or carers of children ages 0-to 5 years;
Record Type: academic journals; grey literature
Date: No limits.

Data Base
MEDLINE (Ovid)
exp family/ OR (parent* OR guardian* OR carer* OR mother* OR father* OR grandparent* OR child* OR prenatal OR pregnan* OR famil*).mp
AND (Financial adj2 (counsel* OR education OR therap* OR service* OR advice OR advisor* OR literacy OR support OR benefit* OR difficul* OR hardship* OR stress* OR pressure* OR problem OR instabilit*) OR (Income maximi* OR income advice OR debt counsel* OR welfare benefit maximi* OR welfare advice* OR money advice worker* OR welfare right* OR local government welfare officer* OR application assistance OR reduc* allocation cost* OR welfare take-up OR debt minimi* OR income management OR reduc* application cost* OR poverty minimi*)).mp.
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AND (MH "Primary Health Care") OR (MH "Secondary Health Care") OR (MH "Tertiary Health Care") OR (MH "Physicians, Family") OR (MH "Physicians") OR (MH "Community Health Workers") OR (MH "Nurses") OR (MH "Midwives") OR (MH "Maternal Health Services") OR (MH "Community Health Nursing+") OR (MH "Family Nursing") OR (MH "Maternal-Child Nursing") OR (MH "Pediatricians") OR (MH "Physicians, Family") OR (MH "Physicians, Women") OR (MH "Physicians") OR (MH "Midwifery") OR (MH "Nurse Midwifery") OR (MH "Patient Navigation") OR (MH "Health Care Delivery, Integrated") OR (MH "Family Centered Care+") OR (MH "Shared Services, Health Care") OR (doctor* OR health nurs* OR clinician* OR "community health worker*" OR "family centred care" OR GPs OR "health personnel" OR "health provider*" OR "healthcare provider*" OR "health-care provider*" OR "health worker*" OR "healthcare worker*" OR "health-care worker*" OR "health service provider*" OR "health service professional*" OR "health professional*" OR "healthcare professional*" OR "health-care professional*" OR "health practitioner*" OR "healthcare practitioner*" OR "health-care practitioner*" OR "health care service" OR "integrated health care delivery" OR nurs* OR "maternal health service*" OR midwi?e* OR "nurse-midwife" OR "patient navigat*" OR physician* OR "primary health care" OR "primary healthcare" OR pediatrician* OR paediatrician* OR "pediatric* care" OR "paediatric* care" OR "primary care provider*" OR practitioner* OR "secondary health care" OR "secondary healthcare" OR "shared services health care" OR "social worker*" OR "secondary care provider*" OR "tertiary health care" OR "tertiary healthcare" OR "tertiary care provider*" OR "patient navigat*").mp
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"application assistance" OR "welfare application assistance" OR "reduc* allocation cost*" OR "welfare take-up" OR "debt minimi*" OR "income management" OR "reduc* application cost*" OR "poverty minimi*" OR "Maternal-Child Welfare" OR "Maternal Welfare" AND doctor* OR "health nurs*" OR clinician* OR "community health worker*" OR "family centred care" OR GPs OR "health personnel" OR "health provider*" OR "healthcare provider*" OR "health-care provider*" OR "health worker*" OR "healthcare worker*" OR "health-care worker*" OR "health service provider*" OR "health service professional*" OR "health professional*" OR "healthcare professional*" OR "health-care professional*" OR "health practitioner*" OR "healthcare practitioner*" OR "health-care practitioner*" OR "health care service" OR "integrated health care delivery" OR nurs* OR "maternal health service*" OR midwi?e* OR "nurse-midwife" OR "patient navigat*" OR physician* OR "primary health care" OR "primary healthcare" OR pediatrician* OR paediatrician* OR "pediatric* care" OR "paediatric* care" OR "primary care provider*" OR practitioner* OR "secondary health care" OR "secondary healthcare" OR "shared services health care" OR "social worker*" OR "secondary care provider*" OR "tertiary health care" OR "tertiary healthcare" OR "tertiary care provider*" OR "patient navigat*"
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Cochrane
family OR parent* OR guardian* OR carer* OR mother* OR father* OR grandparent* OR child* OR prenatal OR pregnan*
AND ''Financial counsel*" OR "financial education" OR "financial therapy" OR "financial service*" OR "financial advice" OR "financial advisor*" OR "financial literacy" OR "financial support" OR "financial benefit*" OR "financial difficul*" OR "financial hardship*" OR "financial stress*" OR "financial pressure*" OR "financial problem*" OR "financial instabilit*" or "Income maximi*" OR "income advice" OR "debt counsel*" OR "welfare benefit maximi*" OR "welfare advice*" OR "money advice worker*" OR "welfare right*" OR "local government welfare officer*" OR "application assistance" OR "welfare application assistance" OR "reduc* allocation cost*" OR "welfare take-up" OR "debt minimi*" OR "income management" OR "reduc* application cost*" OR "poverty minimi*" or "Maternal-Child Welfare" OR "Maternal Welfare"
AND doctor* OR "health nurs*" OR clinician* OR "community health worker*" OR "family centred care" OR GPs OR "health personnel" OR "health provider*" OR "healthcare provider*" OR "health-care provider*" OR "health worker*" OR "healthcare worker*" OR "health-care worker*" OR "health service provider*" OR "health service professional*" OR "health professional*" OR "healthcare professional*" OR "health-care professional*" OR "health practitioner*" OR "healthcare practitioner*" OR "health-care practitioner*" OR "health care service" OR "integrated health care delivery" OR nurs* OR "maternal health service*" OR midwi?e* OR "nurse-midwife" OR "patient navigat*" OR physician* OR "primary health care" OR "primary healthcare" OR pediatrician* OR paediatrician* OR "pediatric* care" OR "paediatric* care" OR "primary care provider*" OR practitioner* OR "secondary health care" OR "secondary healthcare" OR "shared services health care" OR "social worker*" OR "secondary care provider*" OR "tertiary health care" OR "tertiary healthcare" OR "tertiary care provider*" OR "patient navigat*"
Proquest Databases
family OR parent* OR guardian* OR carer* OR mother* OR father* OR grandparent* OR child* OR prenatal OR pregnan* AND "Financial counsel*" OR "financial education" OR "financial therapy" OR "financial service*" OR "financial advice" OR "financial advisor*" OR "financial literacy" OR "financial support" OR "financial benefit*" OR "financial difficul*" OR "financial hardship*" OR "financial stress*" OR "financial pressure*" OR "financial problem*" OR "financial instabilit*" OR "Income maximi*" OR "income advice" OR "debt counsel*" OR "welfare benefit maximi*" OR "welfare advice*" OR "money advice worker*" OR "welfare right*" OR "local government welfare officer*" OR

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"application assistance" OR "welfare application assistance" OR "reduc* allocation cost*" OR "welfare take-up" OR "debt minimi*" OR "income management" OR "reduc* application cost*" OR "poverty minimi*" or "Maternal-Child Welfare" OR "Maternal Welfare"

AND

doctor* OR "health nurs*" OR clinician* OR "community health worker*" OR "family centred care" OR GPs OR "health personnel" OR "health provider*" OR "healthcare provider*" OR "health-care provider*" OR "health worker*" OR "healthcare worker*" OR "health-care worker*" OR "health service provider*" OR "health service professional*" OR "health professional*" OR "healthcare professional*" OR "health-care professional*" OR "health practitioner*" OR "healthcare practitioner*" OR "health-care practitioner*" OR "health care service" OR "integrated health care delivery" OR nurs* OR "maternal health service*" OR midwi?e* OR "nurse-midwife" OR "patient navigat*" OR physician* OR "primary health care" OR "primary healthcare" OR pediatrician* OR paediatrician* OR "pediatric* care" OR "paediatric* care" OR "primary care provider*" OR practitioner* OR "secondary health care" OR "secondary healthcare" OR "shared services health care" OR "social worker*" OR "secondary care provider*" OR "tertiary health care" OR "tertiary healthcare" OR "tertiary care provider*" OR "patient navigat*"

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PRISMA-P checklist

PRISMA-P (Preferred Reporting Items for Systematic review and Meta-Analysis Protocols) 2015 checklist: recommended items to address in a systematic review protocol*

Section and topic	Item No	Checklist item	Page #
ADMINISTRATIVE INFORMATION			
Title:			
Identification	1a	Identify the report as a protocol of a systematic review	2
Update	1b	If the protocol is for an update of a previous systematic review, identify as such	NA
Registration	2	If registered, provide the name of the registry (such as PROSPERO) and registration number	2
Authors:			
Contact	3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	1
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review	8
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	NA
Support:			
Sources	5a	Indicate sources of financial or other support for the review	8
Sponsor	5b	Provide name for the review funder and/or sponsor	8
Role of sponsor or funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol	8
INTRODUCTION			
Rationale	6	Describe the rationale for the review in the context of what is already known	3
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	4
METHODS			
Eligibility criteria	8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	4
Information sources	9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	4,5
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could	4

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		be repeated	Supplementary file 2
Study records:			
Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	5
Selection process	11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	5
Data collection process	11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	5
Data items	12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	5,6
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	5
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	6
Data synthesis	15a	Describe criteria under which study data will be quantitatively synthesised	6
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I^2, Kendall's τ)	6
	15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	NA
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned	6
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)	NA
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	6

NA = Not applicable

From: Shamseer L, Moher D, Clarke M, Ghersi D, Liberati A, Petticrew M, Shekelle P, Stewart L, PRISMA-P Group. Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015: elaboration and explanation. BMJ. 2015 Jan 2;349(jan02 1):g7647.

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